

UNIT I: HTML5 and CSS

Q1. List new tags in HTML5 (any 5 tags with short description)

HTML5 introduced several new semantic and multimedia tags to enhance the structure and functionality of web pages. Below are five important new tags with their short descriptions:

1. **<section>**: Defines a section in a document. It represents a standalone piece of content that has a heading. For example, an article might be broken into sections.
2. **<article>**: Represents self-contained content that can be independently distributed or reused. For example, a blog post, forum post, or news article.
3. **<nav>**: Defines a block of navigation links. Typically used for menus, table of contents, or any set of navigation links.
4. **<figure>**: Used to group media content (like images, charts, or illustrations) with an optional **<figcaption>** to provide a caption.
5. **<audio>**: Allows embedding audio content into a web page. It supports controls like play, pause, and volume.

These tags improve the semantic meaning of a webpage, enhancing readability and accessibility for both developers and browsers.

Q2. Explain table tag with example.

The **<table>** tag in HTML is used to create a table structure to display data in rows and columns. A basic HTML table is constructed using the following tags:

- **<table>**: The container for the table.
- **<tr>**: Defines a table row.
- **<th>**: Defines a header cell in the table (bold and centered by default).
- **<td>**: Defines a standard data cell.

Example:

```
<table border="1">
  <tr>
    <th>Roll No</th>
    <th>Name</th>
  </tr>
  <tr>
    <td>101</td>
    <td>ABC</td>
  </tr>
  <tr>
    <td>102</td>
    <td>XYZ</td>
  </tr>
</table>
```

Explanation:

- The first row is a header row created using **<th>**.
- The next rows are regular data rows using **<td>**.
- The **border="1"** attribute adds a border to the table.

Tables are useful for displaying structured information like schedules, reports, or marksheets.

Q3. What are the different types of Lists in HTML?

1. **Ordered List ()**: Displays items in a specific order (numbered).

```
<ol>
  <li>HTML</li>
  <li>CSS</li>
</ol>
```

2. **Unordered List ():** Displays items in no particular order (bulleted).

```
<ul>
  <li>Apple</li>
  <li>Banana</li>
</ul>
```

3. **Description List (<dl>):** Used for terms and their descriptions.

```
<dl>
  <dt>HTML</dt>
  <dd>Markup Language</dd>
</dl>
```

Q4. Differentiate between HTML and HTML5

Feature	HTML	HTML5
Doctype	Long and complex	Simplified: <!DOCTYPE html>
Multimedia Support	Requires external plugins (like Flash)	Built-in support for <audio> and <video>
Semantic Elements	Limited semantic tags	Introduced tags like <section>, <article>
Form Enhancements	Basic input types	New types like date, email, range, etc.
Graphics Support	No direct support	<canvas> and <svg> added
Storage	Relies on cookies	Supports localStorage and sessionStorage

HTML5 is an upgraded version that provides better structure, multimedia handling, and advanced form controls, making web development more efficient.

Q5. Write short note on ordered list and unordered list. [see Q3]

Q6. What is form? Explain form components with example.

A **form** in HTML is used to collect user input. It sends data to a server using the action and method attributes.

```
<form action="submit.php" method="post">
  <!-- form components -->
</form>
```

Common Form Components:

1. **<input>** – Accepts user input (text, number, checkbox, etc.): `<input type="text" name="username">`
2. **<label>** – Defines a label for inputs: `<label for="username">Username:</label>`
3. **<textarea>** – Multi-line text input: `<textarea name="message"></textarea>`
4. **<select> and <option>** – Dropdown list

```
<select name="gender">
  <option>Male</option>
  <option>Female</option>
</select>
```
5. **<button> Or <input type="submit">** – Used to submit the form

Forms are essential for tasks like login, registration, feedback, and search.

Q7. Explain the attributes of input tag with an example.

The <input> tag is a key element in HTML forms used to take various types of user input. It supports multiple attributes to control behavior and validation.

Common attributes of <input>:

1. **type** – Specifies the input type (text, password, email, etc.): `<input type="email">`
2. **name** – Assigns a name for form submission: `<input name="username">`
3. **value** – Defines default value: `<input value="ABC">`
4. **placeholder** – Displays hint text inside input: `<input placeholder="Enter name">`
5. **required** – Makes the field mandatory: `<input required>`
6. **readonly** – Makes input non-editable: `<input readonly value="Admin">`
7. **maxlength** – Sets character limit: `<input maxlength="10">`

Example: `<input type="text" name="user" placeholder="Enter Username" required>`

This input field accepts text, shows placeholder text, and requires a value before form submission.

Q8. Explain the structure of the HTML5 webpage with an example.

An HTML5 document follows a standard structure that ensures proper rendering across all browsers. It includes semantic elements for better readability and accessibility.

Basic Structure:

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="UTF-8">
    <title>My Webpage</title>
  </head>
  <body>
    <header>
      <h1>Welcome</h1>
    </header>

    <nav>
      <a href="#">Home</a>
      <a href="#">Contact</a>
    </nav>

    <section>
      <h2>About</h2>
      <p>This is a sample HTML5 structure.</p>
    </section>

    <footer>
      <p>© 2025 Ajit Mali</p>
    </footer>
  </body>
</html>
```

Explanation of Key Parts:

- `<!DOCTYPE html>`: Declares HTML5 document.
- `<html>`: Root of the document.
- `<head>`: Contains meta info, title, links.
- `<body>`: Visible content.
- `<header>`, `<nav>`, `<section>`, `<footer>`: Semantic tags for better structure.

This structure improves SEO, accessibility, and maintainability.

Q9. List out limitations of HTML?

While HTML is the backbone of web content, it has several limitations:

1. **No Logic/Programming Capability:** HTML is a markup language, not a programming language. It cannot perform calculations or make decisions (e.g., no loops, conditions).
2. **Static Content Only:** HTML alone cannot create dynamic or interactive content. JavaScript or backend scripting is needed for interactivity.
3. **Poor Data Handling:** HTML cannot connect to databases or manage data. It requires server-side technologies like PHP, Node.js, or Java.
4. **Limited Security:** HTML does not offer security features. Data validation and protection must be handled using JavaScript or server-side code.
5. **Not Suitable for Complex Applications:** For large-scale applications, HTML must be combined with CSS, JavaScript, and frameworks, as it cannot manage app logic or state.
6. **Browser Dependency:** Some HTML tags or features may behave differently in different browsers, especially older ones.

HTML is essential for content structure but is limited in functionality and must be used with other technologies for full-fledged web development.

Q10. What is the purpose of the <datalist> tag in HTML?

The <datalist> tag is used to provide a list of predefined options to an <input> element. It helps users by offering auto-complete suggestions while typing.

Key Features:

- Enhances user input with suggestions.
- Doesn't restrict the user; they can still enter custom values.
- Must be linked to an <input> using the list attribute.

Example:

```
<input list="browsers" name="browser">
<datalist id="browsers">
  <option value="Chrome">
  <option value="Firefox">
  <option value="Edge">
</datalist>
```

Explanation:

- As the user types in the input, the options from the <datalist> appear as suggestions.
- The user can select from suggestions or enter something else.

Use Case: Useful in search bars, dropdowns with flexible inputs, and improving user experience.

Q11. Identify the three ways of color property values which can be specified in CSS?

In CSS, the color property is used to set the text color, and it can be defined in the following three common ways:

1. **Color Names:** Use predefined color names like red, blue, green, etc.:
`p { color: blue; }`
2. **Hexadecimal Values:** Represent colors using hex codes: #RRGGBB:
`h1 { color: #ff0000; } /* red */`
3. **RGB/RGBA Values:** Use rgb(red, green, blue) or rgba (includes alpha for transparency)
`div { color: rgb(0, 128, 0); } /* green */`
`span { color: rgba(255, 0, 0, 0.5); } /* semi-transparent red */`

These methods give flexibility in choosing exact or named colors, and RGBA adds control over transparency.

Q12. Explain types of Style Sheet.

There are **three types of style sheets** in CSS used to apply styles to HTML documents:

1. **Inline Style Sheet:** Applied directly within an HTML tag using the style attribute. Affects only that specific element.

```
<p style="color: red;">Hello</p>
```

2. **Internal (Embedded) Style Sheet:** Written within the <style> tag inside the <head> section of an HTML document. Applies styles to elements within the same HTML file.

```
<style>
  p { font-size: 16px; color: blue; }
</style>
```

3. **External Style Sheet:** Written in a separate .css file and linked to the HTML document using <link>. Useful for applying consistent styles across multiple pages.

```
<link rel="stylesheet" href="style.css">
```

External stylesheets are preferred for large projects due to reusability and maintainability, while inline styles are used for quick, single-element changes.

Q13. Explain uses of Classes in CSS with suitable example.

In CSS, a **class selector** is used to apply the same style to multiple elements by assigning them a common class name.

Key Points:

- Defined using a dot . followed by the class name in CSS.
- Added to HTML elements using the class attribute.
- Allows styling multiple elements consistently and flexibly.

Example:

```
<style>
  .highlight {
    color: white;
    background-color: navy;
    padding: 5px;
  }
</style>

<p class="highlight">This paragraph is highlighted.</p>
<div class="highlight">This div is also highlighted.</div>
```

Explanation:

- The .highlight class defines a set of styles.
- Both <p> and <div> use the same class to share those styles.
- This helps reduce code repetition and improves maintainability.

Use Case: Useful for applying consistent styles like buttons, alerts, or layout blocks across multiple parts of a webpage.

Q14. Explain embedded style sheet with example.

An **embedded style sheet** (also called **internal CSS**) is written inside the <style> tag within the <head> section of an HTML document. It allows you to define CSS rules that apply to the current page only.

Syntax:

```
<head>
  <style>
    h1 {
      color: green;
      text-align: center;
    }
    p {
      font-size: 14px;
      color: gray;
    }
  </style>
</head>
```

Explanation:

- The `<style>` tag wraps the CSS rules.
- The styles apply to all `<h1>` and `<p>` elements in the same HTML page.
- No external file is needed.

Use Case: Embedded styles are useful when:

- You want to style a single page quickly.
- You don't want to maintain a separate CSS file.

Limitation: Not reusable across multiple pages like external stylesheets.

Q15. Differentiate between Inline and External Style Sheet

Feature	Inline Style Sheet	External Style Sheet
Location	Inside HTML element using style attribute	In a separate .css file
Syntax	<code><p style="color:blue;">Text</p></code>	<code><link rel="stylesheet" href="style.css"></code>
Reusability	Cannot be reused	Can be reused across multiple pages
Maintenance	Difficult to maintain for large pages	Easy to maintain and update in one place
Performance	Slower if styles are repeated multiple times	Better performance for multi-page websites

Summary:

- **Inline** is quick but messy.
- **External** is clean, reusable, and ideal for large or multi-page projects.

Q6. What is selector class? Explain the pseudo class selector in CSS with example.

Class Selector: A **class selector** selects HTML elements with a specific class attribute. It is defined using a dot (.) followed by the class name.

```
.note {
  color: green;
}
<p class="note">This is important.</p>
```

Pseudo-Class Selector: A **pseudo-class** is used to define a special state of an element, such as when a user hovers over a link or focuses on an input field.

Common pseudo-classes:

- `:hover` – applies when the user hovers over an element.
- `:focus` – applies when an element (like input) is focused.
- `:first-child` – targets the first child element.

```
a:hover {
  color: red;
}
input:focus {
  border: 2px solid blue;
}
```

Use Case: Pseudo-classes improve interactivity and user experience without JavaScript.

Q17. What are different parts of CSS?

A CSS rule is made up of several parts that define how elements should be styled. The main components are:

1. **Selector:** Identifies the HTML element(s) to be styled. Example: p, .class, #id
2. **Declaration Block:** Enclosed in {} and contains one or more **declarations**.
3. **Declaration:** Consists of a **property** and a **value**, separated by a colon. Each declaration ends with a semicolon.

```
h1 {
  color: blue;
  font-size: 20px;
}
```

- **Selector:** h1
- **Declaration Block:** { color: blue; font-size: 20px; }
- **Declarations:** color: blue; and font-size: 20px;
- **Properties:** color, font-size
- **Values:** blue, 20px

These parts together define how the selected element should appear on the webpage.

Q18. What are CSS backgrounds, list the properties?

CSS allows you to control the background of elements using various **background properties**. These properties define color, images, position, and repetition behavior of backgrounds.

Common Background Properties:

1. **background-color:** Sets the background color of an element: `background-color: lightblue;`
 2. **background-image:** Sets an image as the background: `background-image: url('bg.jpg');`
 3. **background-repeat:** Controls if/how the background image is repeated. Values: repeat, no-repeat, repeat-x, repeat-y: `background-repeat: no-repeat;`
 4. **background-position:** Sets the starting position of the background image: `background-position: center top;`
 5. **background-size:** Scales the background image: `background-size: cover;`
 6. **background-attachment:** Defines if background scrolls/ fixed: `background-attachment: fixed;`
 7. **background (shorthand):** `background: url('bg.jpg') no-repeat center/cover fixed;`
-

Q19. What is the difference between class and id selector?

Feature	Class Selector	ID Selector
Syntax	Uses dot (.) before name	Uses hash (#) before name
Usage	Can be applied to multiple elements	Should be unique for a single element
Example (CSS)	<code>.btn { color: blue; }</code>	<code>#header { background: gray; }</code>

Example (HTML)	<code><div class="btn">Click</div></code>	<code><div id="header">Top</div></code>
Purpose	General styling for reusable elements	Specific styling for a unique element
Reusability	Highly reusable	Not reusable (meant for one-time use)

Summary:

- **Class** is ideal for shared styles.
- **ID** is for uniquely identifying and styling one element.

Q20. Briefly explain CSS Syntax and Style.

CSS Syntax: A CSS rule consists of a **selector** and a **declaration block**.

```
selector {
  property: value;
}
```

Example:

```
p {
  color: blue;
  font-size: 16px;
}
```

- **Selector:** p targets all <p> elements.
- **Properties:** color, font-size
- **Values:** blue, 16px
- **Declaration Block:** Enclosed in { }, contains one or more property-value pairs.

CSS Styling Features:

- Controls layout, colors, fonts, spacing, and more.
- Allows consistent design across multiple pages.
- Can be written inline, internally, or externally.

Types of Units:

- **Length:** px, em, %
- **Color:** name, hex, RGB
- **URL:** for background images, fonts, etc.

CSS separates content (HTML) from presentation, making web design more efficient and maintainable.